

PROJECT REPORT

National Air Quality Index (AQI) Analytics & Engineering Pipeline

An end-to-end SQL Server data warehouse, ETL pipeline, and Power BI decision-support dashboard built on 550+ air-quality monitoring stations across India (2009–2024)

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Data Source: Kaggle (Central Pollution Control Board monitoring network)

1. Executive Summary

This project builds a production-style analytics pipeline that transforms raw, inconsistent air-quality sensor readings from over 550 monitoring stations across India into a governed, decision-ready dataset. The pipeline ingests 15-minute pollutant readings spanning 2009–2024, structures them through a three-layer (Bronze → Silver → Gold) SQL Server warehouse, and surfaces the results through a 3-page Power BI dashboard covering forecasting, seasonal/weekly trend analysis, and meteorological correlation.

The end goal was not just to store the data, but to make it usable for a business question: where and when does air quality deteriorate, and what conditions drive it. The resulting dashboard identifies clear seasonal windows (Q4 pollution spikes), weekly patterns tied to commercial activity, and consistent geographic hotspots — insights that map directly onto real policy and operational decisions (e.g., timing of emission controls, targeted monitoring of specific stations).

2. Problem Statement & Objective

Public air-quality datasets are typically distributed as thousands of loosely structured CSV files, one per station, with inconsistent columns, missing readings, and no computed Air Quality Index (AQI). This makes them unusable for direct analysis or reporting. The objective of this project was to:

- Consolidate raw multi-station sensor data into a single, query-ready warehouse.
- Apply a layered transformation strategy so raw, cleaned, and business-ready data are each auditable and separately queryable.
- Compute a standardized daily AQI per station from individual pollutant concentrations.
- Deliver the results as an interactive dashboard that supports trend analysis, forecasting, and root-cause exploration (meteorological drivers).

3. Data Source & Scope

3.1 Source

Raw data was sourced from Kaggle, covering 550+ Central Pollution Control Board (CPCB) monitoring stations distributed across Indian states, with readings recorded at 15-minute intervals from 2009 to 2024 — amounting to roughly 83 million+ raw records across 19+ state-wise directories of CSV files.

3.2 Structure

Each station's data arrived as a separate CSV file, nested inside state-level sub-folders. Column sets were not standardized: a station file could contain any subset of the following 20 fields, but never fields outside this set:

pm2.5, pm10, no, no2, so2, co, nox, ozone, at, rh, ws, vws, wd, bp, timestamp, nh3, benzene, eth_benzene, toluene, mp_xylene, xylene

This column inconsistency was the central data-quality challenge the ingestion pipeline had to solve before any analysis could begin.

4. Ingestion Pipeline (Python → SQL Server)

A Python-based extraction layer was built to traverse the nested folder structure, standardize each file, and load everything into a single staging table in Microsoft SQL Server.

- **Connectivity:** pyodbc was used to establish the connection between Python and MS SQL Server.
- **Discovery:** os and glob were used to walk the main directory and each state sub-folder, collecting the full file path of every station CSV (excluding the separate stations.csv metadata file) into a single processing list.
- **Standardization:** Each file was reindexed against the full 20-column schema using pandas so that every table had a uniform column structure regardless of which fields the original file contained; missing values were filled as NaN, and a station_code column was added to trace every row back to its source station.
- **Type strategy:** All columns were loaded as NVARCHAR at this stage — correctness of type was deferred to the transformation layer rather than enforced during ingestion, keeping the load step fast and resilient to malformed source data.
- **Load:** All standardized station files were combined and loaded via a bulk insert query into one consolidated raw table, avoiding the overhead of thousands of individual row-by-row inserts.

Libraries used: pyodbc, os, glob, pandas.

5. Database Design & Performance Optimization

5.1 Schema

Table	Grain	Description
data (readings)	1 row per station per 15-min interval	Core fact table holding every pollutant and meteorological reading, 2009–2024.
stations	1 row per station	Dimension table with geographic metadata (state, location, coordinates) per station.

5.2 Columnstore Indexing

Given the fact table's scale (15-minute granularity across 550+ stations over 15 years), a columnstore index was applied to the main readings table. This was chosen deliberately over a standard row-store index because analytical queries in this project are read-heavy and column-scoped (e.g., aggregating pm2.5 across millions of rows) rather than single-row lookups — exactly the access pattern columnstore compression is designed for.

Metric	Impact of Columnstore Index
Query execution time	50–60% faster
Storage footprint	Reduced by up to 90–95%

6. Medallion Architecture: Bronze → Silver → Gold

To keep the pipeline auditable and business-ready, transformations were organized into three progressive layers rather than done in one step — each layer has a clear, single responsibility, so raw data, cleaned data, and reporting-ready data can each be inspected independently.

Layer	Responsibility	Key Transformations
Bronze	Raw ingestion	Ingests standardized CSVs as-is (NVARCHAR); columnstore index applied here too, for fast bulk insertion and efficient storage of raw volume.
Silver	Cleaned & typed	Casts columns from VARCHAR to correct data types; computes rolling averages (8-hour for ozone, 1-hour for CO, matching standard pollutant-specific measurement windows); computes daily averages and daily maximum concentrations for other pollutants; removes records for unknown/unmapped stations; columnstore index applied for query and storage efficiency.
Gold	Business-ready	Calculates daily AQI per station from individual pollutant sub-indices; caps AQI at the standard maximum of 500 for extreme readings; renames columns to clear, business-friendly labels for downstream reporting.

This layered approach mirrors how enterprise analytics teams structure warehouses: it isolates data-quality logic (Silver) from ingestion (Bronze) and business logic (Gold), so any single layer can be re-run, debugged, or extended without touching the others.

7. Power BI Dashboard

The Gold layer feeds a 3-page Power BI dashboard designed around three distinct analytical needs: a real-time/forecast snapshot per station, historical seasonal and weekly trend analysis, and root-cause exploration via meteorological correlation.

7.1 Page 1 — Station Snapshot & 7-Day AQI Forecast

This page gives a station-level operational view: current AQI status with severity classification (e.g., “Unhealthy”), live pollutant breakdown (PM10, PM2.5, CO, Ozone, NO2, SO2, NH3), meteorological context (humidity, wind, pressure), and a 7-day forward AQI and temperature outlook, filterable by station and date.

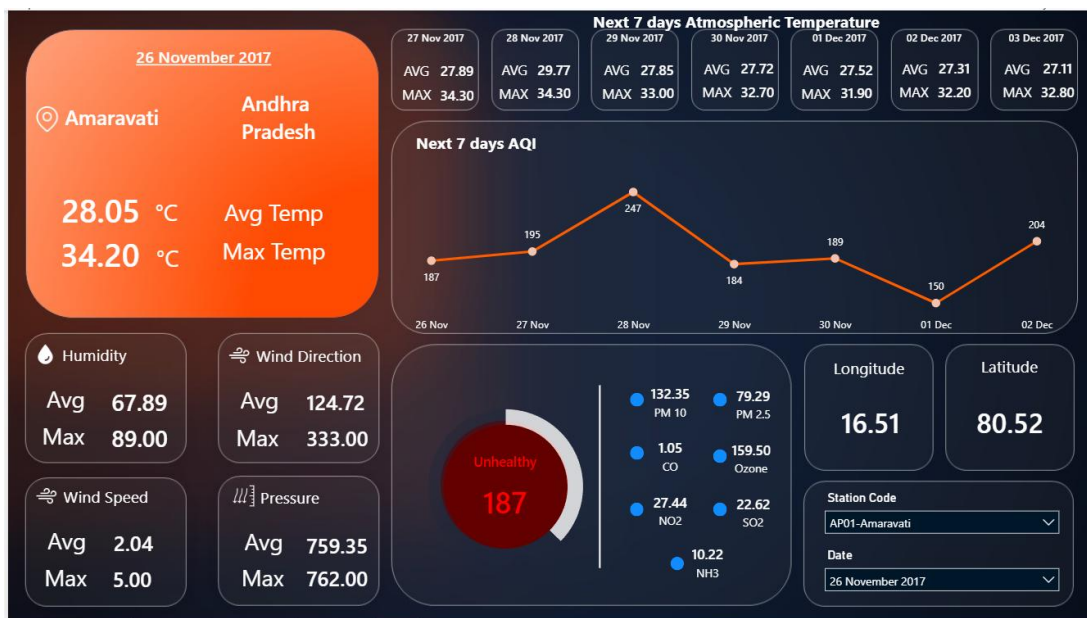


Figure 1 — Station snapshot and 7-day AQI/temperature forecast (Amaravati, Andhra Pradesh shown).

7.2 Page 2 — Seasonal & Weekly Trend Analysis

This page aggregates AQI by month and by day of the week, alongside the top 5 best- and worst-performing stations for the selected year, filterable by state and station code.



Figure 2 — Monthly and day-of-week AQI trends, with top 5 best/worst stations (Bihar / BR03-Muzaffarpur, 2015 shown).

- **Seasonal spike:** AQI rises sharply through Q4, climbing from ~225 in September to ~394 by December in high-pollution hubs such as Bihar and Delhi — consistent with winter temperature inversion, seasonal weather patterns, and agricultural (stubble) burning.
- **Weekly pattern:** AQI peaks mid-week (~381 on Tuesday) and drops on weekends (~273 on Sunday), pointing to commercial vehicular and industrial activity as a significant contributor.

7.3 Page 3 — Meteorological Correlation Analysis

This page tests AQI against key meteorological variables using scatter plots with trend lines, alongside a multi-year AQI trend and historical best/worst station rankings.



Figure 3 — AQI correlation with wind speed, humidity, barometric pressure, and vertical wind shear; multi-year AQI trend and station rankings.

- **Wind speed:** shows a clear negative correlation with AQI — higher wind speeds disperse particulate concentration and rapidly improve air quality.
- **Barometric pressure:** shows a positive correlation with AQI, consistent with high-pressure systems trapping pollutants closer to the surface (temperature inversion).
- **Hotspots:** Stations in Bihar (BR03-Muzaffarpur), Delhi (DL33, DL09, DL30), and Andhra Pradesh consistently register the worst annual AQI, while coastal and southern stations (e.g., Chennai, Hyderabad, Mumbai) maintain a significantly lower baseline year over year.

8. Why This Matters for Decision-Making

Beyond the technical build, the pipeline was designed to answer decision-relevant questions rather than just present data:

- **Timing decisions:** the clear Q4 seasonal spike gives a concrete window for pre-emptive intervention (e.g., stricter emission controls before September rather than reactive measures in December).
- **Operational decisions:** the weekday/weekend gap suggests vehicular and industrial activity restrictions would have more impact if targeted at mid-week peaks rather than applied uniformly.
- **Resource allocation:** consistent multi-year hotspots (Muzaffarpur, Delhi stations) identify where monitoring infrastructure and mitigation budgets deliver the most impact.
- **Engineering trade-offs:** the columnstore indexing decision was validated with measured before/after impact (50–60% query speed, 90–95% storage reduction) rather than assumed — a habit of quantifying infrastructure decisions rather than treating them as default choices.

9. Conclusion & Future Scope

This project demonstrates a complete analytics workflow: handling messy, inconsistent real-world data at scale, structuring it through a disciplined layered warehouse, making deliberate and measured performance trade-offs, and translating the output into a dashboard built around specific, answerable business questions rather than generic charts.

Potential extensions include: automating the Bronze→Silver→Gold refresh as a scheduled pipeline, adding a predictive model for AQI forecasting (the current forecast view is trend-based) using historical + meteorological features, and expanding correlation analysis to include additional pollutant precursors such as traffic density or industrial activity indices where available.